

# Harshitha Palaram

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## EDUCATION

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|-----------------------------------------------------------|---------------------|
| <b>Shiv Nadar University</b> , Chennai                    | Aug 2023 – May 2027 |
| <i>B.Tech in Artificial Intelligence and Data Science</i> | CGPA: 7.713         |
| <b>FIITJEE Junior College</b> , Hyderabad                 | May 2021 – May 2023 |
| <i>Intermediate (Class XII)</i>                           | 95.4%               |
| <b>Royale Concorde International School</b> , Bengaluru   | Till May 2021       |
| <i>Class X (CBSE)</i>                                     | 96.6%               |

## EXPERIENCE

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| <b>Machine Learning Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Apr 2026 – May 2026 |
| <i>Cognifyz IT Solutions Pvt. Ltd.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Remote              |
| <ul style="list-style-type: none"><li>Engineered and evaluated end to end <b>machine learning pipelines</b> for <b>restaurant rating prediction, recommendation systems, cuisine classification, and location analytics</b> using <b>Python, Pandas, NumPy, Scikit-learn, TF-IDF, and cosine similarity</b>, strengthening predictive modeling and data analysis workflows. <a href="#">[Certificate]</a></li><li>Optimized regression models using <b>Random Forest, XGBoost, and CatBoost</b>, achieving <b>95.35% R<sup>2</sup></b> with <b>0.33 RMSE</b>; leveraged <b>DBSCAN clustering</b> and <b>SHAP explainability</b> to extract actionable insights and improve model interpretability.</li></ul> |                     |

## PROJECTS

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| <b>Glacial Lake Outburst Flood Detection (GLOFeagles'26)</b>   <i>U-Net++</i> , <i>EfficientNet-B4</i>   <a href="#">GitHub</a>                                                                                                                                                                                                                                                                                                                                                | 2026 |
| <ul style="list-style-type: none"><li>Architected a dual-task deep learning pipeline using <b>EfficientNet-B4</b> for <b>6-class classification</b> and <b>U-Net++</b> for semantic segmentation of glacial lakes from satellite imagery.</li><li>Achieved <b>0.6260 Mean IoU</b>, <b>0.6920 Dice/F1</b>, and <b>0.7656 Precision</b> on <b>2,220 image-mask pairs</b> using hybrid <b>Focal + Tversky Loss</b>, mixed precision training (AMP), and Albumentations.</li></ul> |      |
| <b>NTIRE 2026 Rip Current Detection &amp; Segmentation Challenge</b>   <i>PyTorch</i> , <i>YOLOv8</i> , <i>OpenCV</i>   <a href="#">GitHub</a>                                                                                                                                                                                                                                                                                                                                 | 2026 |
| <ul style="list-style-type: none"><li>Designed a YOLOv8-based computer vision pipeline for rip current detection using segmentation, contour extraction, morphological operations, and post-processing for accurate localization.</li><li>Implemented multi-scale inference and test-time augmentation (TTA), contributing to a <b>Top 7 global ranking among 900+ submissions</b> at the NTIRE CVPR 2026 Workshop Challenge.</li></ul>                                        |      |
| <b>FinAscend: AI-Powered Financial Decision System</b>   <i>Monte Carlo Simulation</i> , <i>Explainable AI</i>   <a href="#">GitHub</a>                                                                                                                                                                                                                                                                                                                                        | 2026 |
| <ul style="list-style-type: none"><li>Developed an AI powered financial analytics platform integrating OCR and API pipelines to process invoices and financial statements for small businesses.</li><li>Implemented <b>Monte Carlo simulations</b>, probabilistic risk forecasting, and Explainable AI techniques to predict liquidity, runway, and bankruptcy risk.</li></ul>                                                                                                 |      |
| <b>Learn-Lynx: GenAI-Powered Study Partner</b>   <i>LangChain</i> , <i>Hugging Face</i> , <i>RAG</i> , <i>Streamlit</i>   <a href="#">GitHub</a>                                                                                                                                                                                                                                                                                                                               | 2025 |
| <ul style="list-style-type: none"><li>Created a Retrieval-Augmented Generation (RAG) study assistant using <b>LangChain</b>, <b>Sentence Transformers</b>, and Hugging Face for semantic search, summarization, and question answering.</li><li>Delivered an interactive Streamlit application providing personalized learning plans and real-time AI-assisted study support through vector-based document retrieval.</li></ul>                                                |      |
| <b>Food Delivery Web Application</b>   <i>React.js</i> , <i>Node.js</i> , <i>Express.js</i> , <i>MongoDB</i>   <a href="#">GitHub</a>   <a href="#">Live Demo</a>   <a href="#">Admin</a>                                                                                                                                                                                                                                                                                      | 2025 |
| <ul style="list-style-type: none"><li>Built a full-stack food ordering platform using <b>React.js</b>, <b>Node.js</b>, <b>Express.js</b>, and <b>MongoDB</b> featuring authentication, cart management, and order processing.</li><li>Implemented <b>JWT authentication</b>, role-based access control, RESTful APIs, and deployed production-ready applications on <b>Vercel</b> and <b>Render</b>.</li></ul>                                                                 |      |

## TECHNICAL SKILLS

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| <b>Languages:</b> Python, Java, C, JavaScript (ES6+), SQL, HTML/CSS                                                                           |
| <b>Frameworks &amp; Web:</b> React.js, Node.js, Express.js, FastAPI, REST APIs, JWT, Tailwind CSS                                             |
| <b>AI/ML:</b> PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLOv8, Hugging Face Transformers, Sentence Transformers, RAG, Semantic Segmentation |
| <b>Libraries:</b> NumPy, Pandas, Matplotlib, NLTK                                                                                             |
| <b>Databases:</b> MongoDB, MySQL                                                                                                              |
| <b>Tools:</b> Git, GitHub, Docker, Linux, Jupyter Notebook, VS Code, Vercel, Render, Streamlit                                                |

## ACHIEVEMENTS

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| <b>Top 5 Finalist – GLOFeagles Challenge (NCVPRIPG 2026)</b>                                                                                                                      | 2026 |
| <ul style="list-style-type: none"><li>Selected among the <b>Top 5 teams</b> and invited to present the solution at the <b>NCVPRIPG 2026</b> finale in LNMIIT , Jaipur.</li></ul>  |      |
| <b>Top 7 Global Rank NTIRE 2026 Rip Current Detection &amp; Segmentation Challenge (CVPR)</b>                                                                                     | 2026 |
| <ul style="list-style-type: none"><li>Secured <b>7th place among 900+ global submissions</b> in an international computer vision challenge. <a href="#">Leaderboard</a></li></ul> |      |
| <b>Finalist – SNUC Hacks'26 FinTech Hackathon</b>                                                                                                                                 | 2026 |
| <ul style="list-style-type: none"><li>Presented <b>FinAscend</b>, an AI-powered financial decision support system, and advanced to the hackathon finals.</li></ul>                |      |

## CERTIFICATIONS

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| <b>Affective Computing</b>   <i>NPTEL</i>   <a href="#">Certificate</a>                       | Jan 2026 – Apr 2026 |
| <b>Programming in Java</b>   <i>NPTEL</i>   <a href="#">Certificate</a>                       | Jul 2025 – Oct 2025 |
| <b>Machine Learning Specialization</b>   <i>DeepLearning.AI</i>   <a href="#">Certificate</a> | Feb 2025 – Apr 2025 |